In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

A curve C has equation

$$x = \sin^2 4y \qquad 0 \leqslant y \leqslant \frac{\pi}{8} \qquad 0 \leqslant x \leqslant 1$$

The point P with x coordinate $\frac{1}{4}$ lies on C

(a) Find the exact y coordinate of P

(2)

(b) Find $\frac{\mathrm{d}x}{\mathrm{d}y}$

(2)

(c) Hence show that $\frac{\mathrm{d}y}{\mathrm{d}x}$ can be written in the form

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{1}{\sqrt{q + r(x + s)^2}}$$

where q, r and s are constants to be found.

(3)

Using the answer to part (c),

(d) (i) state the x coordinate of the point where the value of $\frac{\mathrm{d}y}{\mathrm{d}x}$ is a minimum,

(ii) state the value of $\frac{\mathrm{d}y}{\mathrm{d}x}$ at this point.

(2)